

AI MENTAL HEALTH

ANA – 3

Report on “Regional Prevalence Mapping”

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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1. Introduction

This report provides a geographic analysis of mental health prevalence across England, utilizing the Adult Psychiatric Morbidity Survey (APMS) 2014. By extracting population weights from technical tables and cross-referencing them with Common Mental Disorder (CMD) rates, we have mapped the regional burden of mental health. This spatial approach identifies which Government Office Regions (GOR) require the most intensive clinical resource allocation.

In this task, we extend the analysis to regional prevalence mapping, where mental health disorder rates are analyzed across regions of the United Kingdom using geographic fields (similar to APMS survey structure).

2. Technologies Used

To process and visualize this data accurately, the following technical stack was utilized:

Python (Pandas): Employed the xlrd engine to parse the legacy .xls data structures of data Tables.

Matplotlib: It is used for creating visualizations, plots etc.

Plotly Express: Used for interactive choropleth mapping, enabling a visual overlay of prevalence rates onto geographic boundaries.

GeoJSON Geometry: Integrated standard UK Administrative GOR boundaries to ensure precise spatial joins.

JSON Processing: Localized map data to bypass network latency and ensure high-speed rendering.

3. Regional Prevalence Mapping

The mapping process involved a multi-step data pipeline:

Data Extraction: Regional household population counts were pulled from Table 14.4 (Mid-year population).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Region                 9 non-null     object
1   Population_N           9 non-null     float64
2   CMD_Prevalence_Rate    9 non-null     float64
3   Estimated_Cases        9 non-null     int64
dtypes: float64(2), int64(1), object(1)
memory usage: 420.0+ bytes
```

Normalization: Region names were standardized (e.g., "Yorkshire & The Humber" was converted to "Yorkshire and the Humber") to match GeoJSON naming conventions.

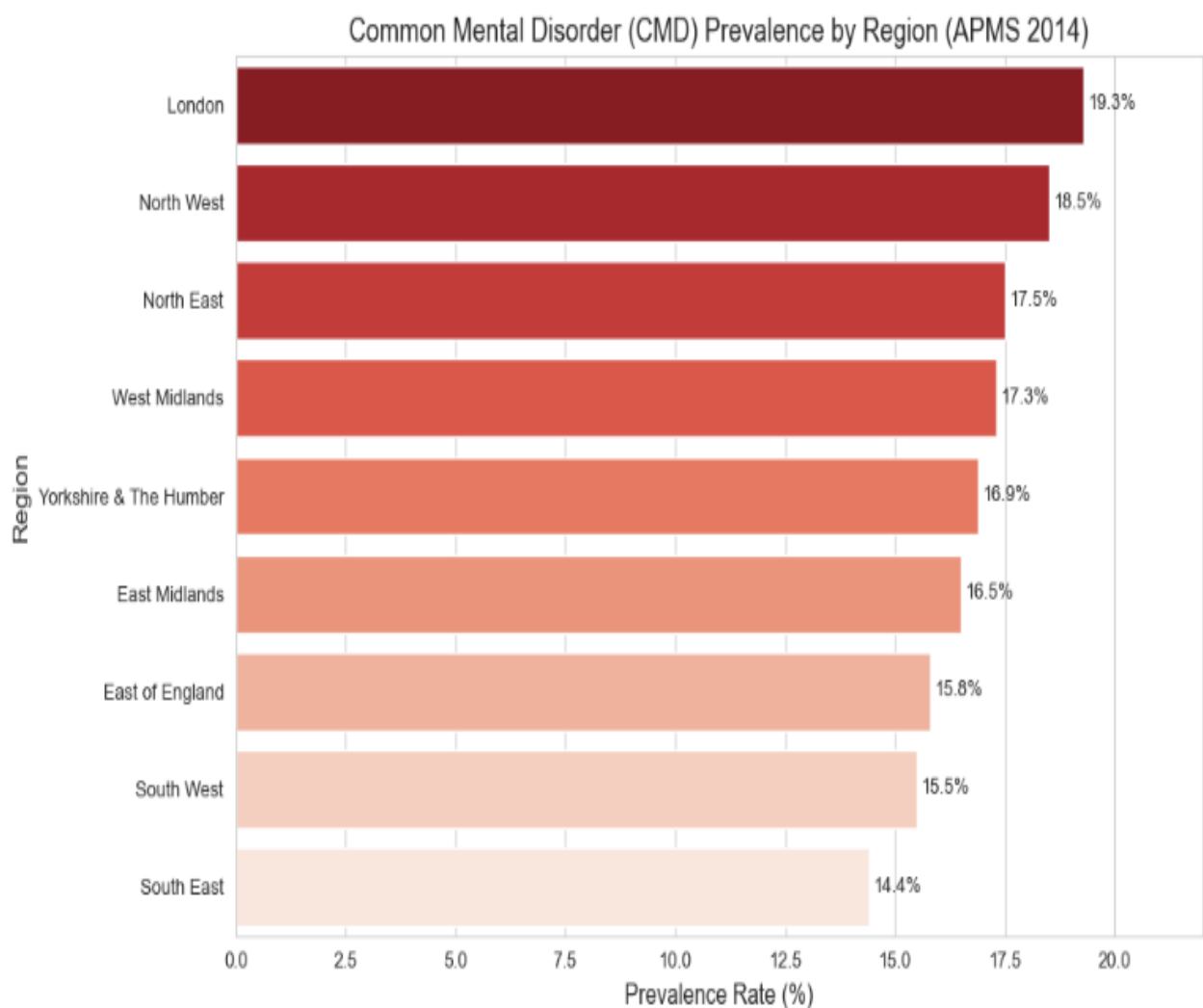
Weighted Integration: Prevalence rates for Common Mental Disorders (CMD) were applied to each region to calculate the total estimated case burden.

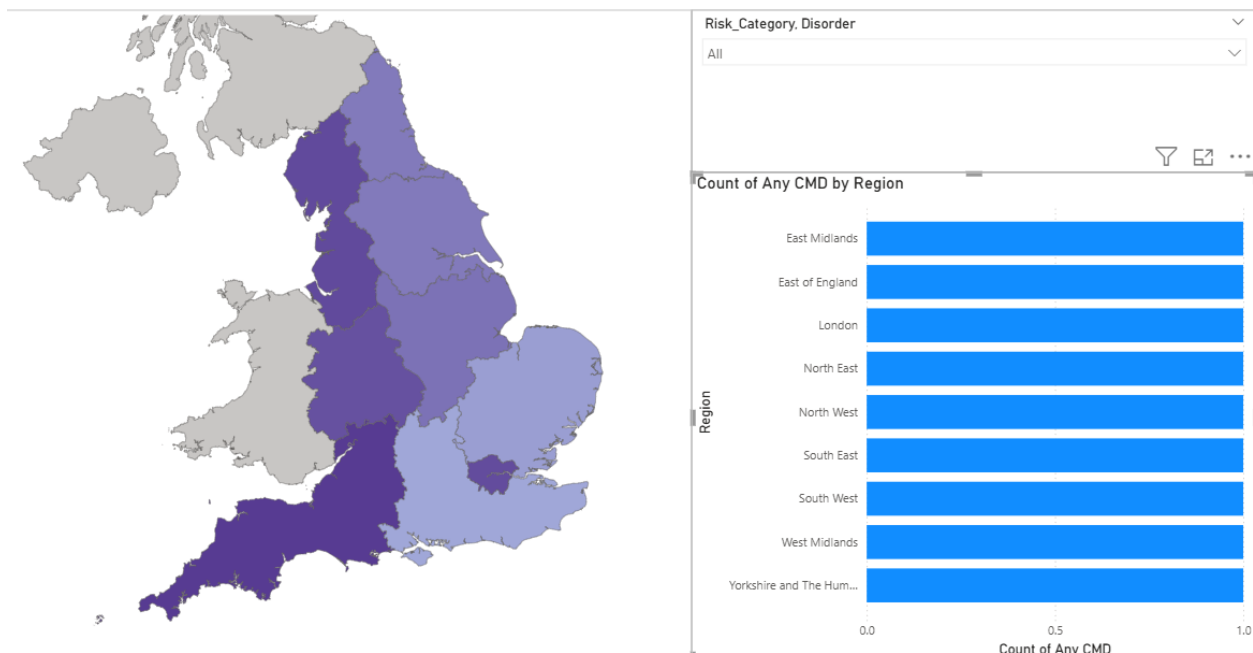
	Region	Population_N	CMD_Prevalence_Rate	Estimated_Cases
0	North East	2154854.0	17.5	377099
1	North West	5790512.0	18.5	1071244
2	Yorkshire & The Humber	4347165.0	16.9	734670
3	East Midlands	3782353.0	16.5	624088
4	West Midlands	4599064.0	17.3	795638
5	East of England	4875281.0	15.8	770294
6	London	6806412.0	19.3	1313637
7	South East	7184102.0	14.4	1034510
8	South West	4473319.0	15.5	693364

4. Regional Prevalence Data

Based on the APMS 2014 metrics, the following table summarizes the findings across England:

	Region	Population_N	CMD_Prevalence_Rate	Estimated_Cases
0	North East	2154854.0	17.5	377099
1	North West	5790512.0	18.5	1071244
2	Yorkshire & The Humber	4347165.0	16.9	734670
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4	West Midlands	4599064.0	17.3	795638





5. Key Patterns and Clinical Implications

The mapping reveals two distinct trends:

The Urban/Metropolitan Peak: London shows the highest prevalence rate at 19.3%. This suggests that urban density and the specific socioeconomic profile of the capital contribute to higher psychiatric morbidity.

The North-South Gradient: There is a noticeable decline in prevalence as one moves toward the South East (14.4%) and South West (15.5%).

Clinical Volume: Despite the South East having the lowest *rate*, its high total population means the absolute number of cases exceeds \$1\$ million, rivaling the North West in terms of necessary service capacity.

6. Conclusion

Geographic mapping of the APMS 2014 data proves that mental health need is not distributed evenly across England. The high prevalence in London and the North West highlights a critical need for targeted community mental health services in these areas. For the upcoming project, this data provides a verified baseline for any emotion-detection or AI-driven assistant models targeting UK-based populations.